Explaining is Harder Than Predicting Alone

Evaluating Concept-based Explanations of MLLMs as ICL Visual Classifiers

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MAIN FINDING

Forcing MLLMs to produce concept-based explanations alongside their predictions degrades accuracy *monotonically* — from **93.8%** with no explanation to **90.1%** under Description-Logics axioms. Classification accuracy *alone* does not reflect explanation quality, but when models succeed, **Local Discriminativeness** of the explanation is the strongest predictor of correct predictions.

01 The ICL (In-Context Learning) *Explain — Classify — Evaluate* pipeline

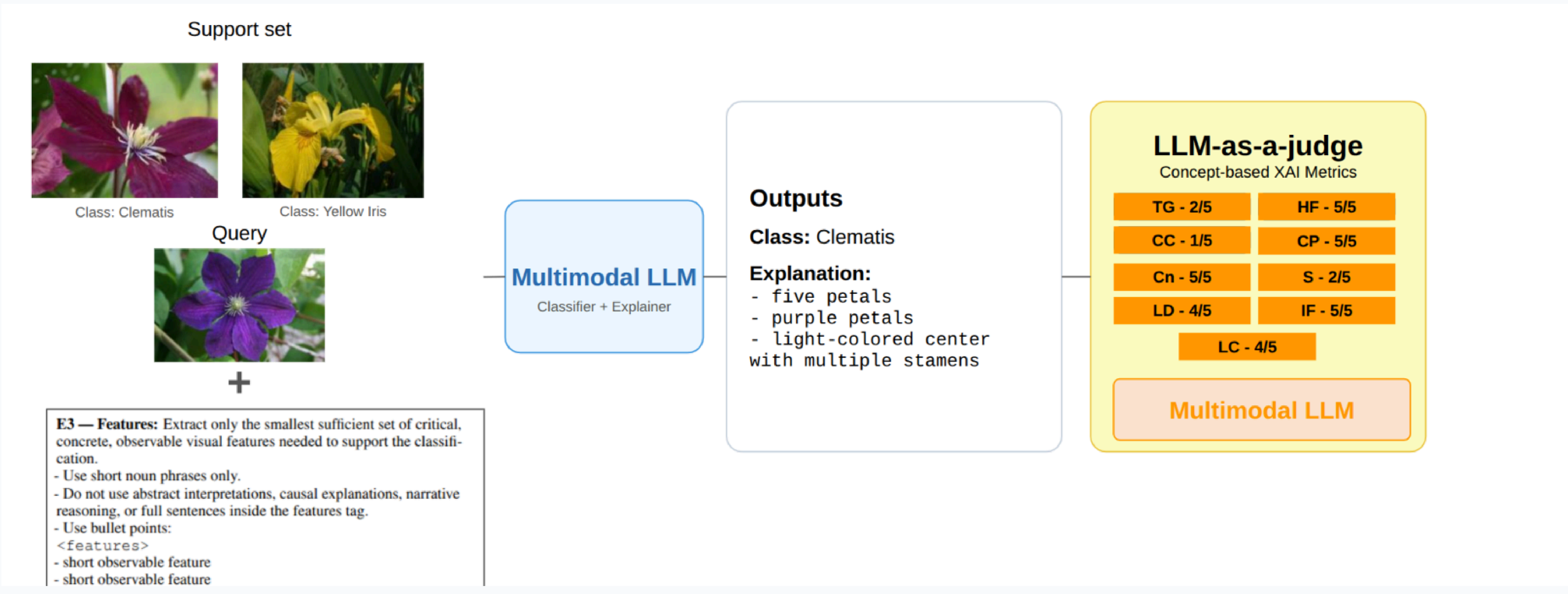

Figure 1. Overview of the In-Context Learning *Explain–Classify–Evaluate* pipeline. One **Multimodal LLM** acts as both classifier and explainer; a separate **LLM-as-a-judge** evaluates nine concept-based XAI metrics (1–5 ↑) in a zero-shot setting.

1. **Few-shot ICL prompt.** N labelled support images + a query image are placed in the context window, together with one of the five explanation conditions E1–E5.
2. **Explain + Classify.** The MLLM emits a class label and a structured explanation (here under E3 — a minimal set of observable visual features).
3. **Evaluate.** An independent judge scores the explanation on the nine XAI metrics defined in Table 1.

02 Motivation

Few-shot ICL lets **MLLMs** classify images from a handful of examples, but *how* they use that context stays opaque. CoT traces may not reflect actual internal computation, so we probe whether MLLMs can produce **concept-based explanations**: human-readable and machine-verifiable statements grounded in the visual evidence.

03 Method — Five explanation conditions

N-way K-shot ICL ($N \in \{2,3,4\}$, $K \in \{1,5\}$). The model predicts *and* explains under one of five conditions of increasing semantic complexity.

- **E1** Classification only *(baseline)*
- **E2** Natural-Language Explanation (NLE)
- **E3** Features-based explanation
- **E4** Feature-value pairs explanation
- **E5** Description-Logics (DL) axioms explanation

04 Evaluation — LLM-as-a-judge

An **independent MLLM judge** (`gpt-5-thinking-mini` , zero-shot) scores every output against the query image and the rubric across the **nine concept-based XAI metrics we introduce** (1–5) — a core contribution of this work.

TABLE 1 · NINE CONCEPT-BASED CONTRIBUTED XAI METRICS (1 = WORST, 5 = BEST ↑)

TG	Textual Groundedness	All salient image concepts are mentioned in the explanation.
HF	Hallucination Free	Every claim is visually verifiable in the image.
CC	Concept Counting	Concept counts match the image exactly.
Cp	Comprehensibility	Readable and accessible to non-expert users.
Cn	Conciseness	Conveys only necessary information without redundancy.
S	Specificity	Relies on precise, non-generic details of the sample.
LD	Local Discriminativeness	Highlights features distinguishing the predicted class from others.
IF	Instruction Following	Complies with all explicit format and structure constraints.
LC	Logical Coherence	Sentences build into a smooth, valid deduction.

05 Experimental set-up

Datasets. CIFAR-10 · DTD · Oxford Flowers 102 · Oxford-IIIT Pets

Models (Temperature = 0).

Gemini 2.5 Flash · *Gemma 4 26B* · *Qwen3-VL 8B* · *LLaMA 4 Scout*

2,080 runs total: 26 trials × 4 models × 4 datasets × 5 conditions. Support sets fixed across models and conditions (seed = 42) for direct comparability.

06 Results — Accuracy drops with explanation rigour

E1	E2	E3	E4	E5
93.8% baseline	**93.6**%	**93.1**%	**92.3**%	**90.1**%

Mean accuracy across 4 models × 4 datasets × 6 few-shot configs (104 obs / cell). Drop is sharpest for Qwen3 VL 8B (**95.1** → **83.0%**); Gemini 2.5 Flash stays above **95%** across all conditions.

TAKE-AWAY Deciding and explaining simultaneously is harder than deciding alone. Demanding richer explanations imposes a measurable cost on predictive accuracy.

07 Explanation quality across conditions

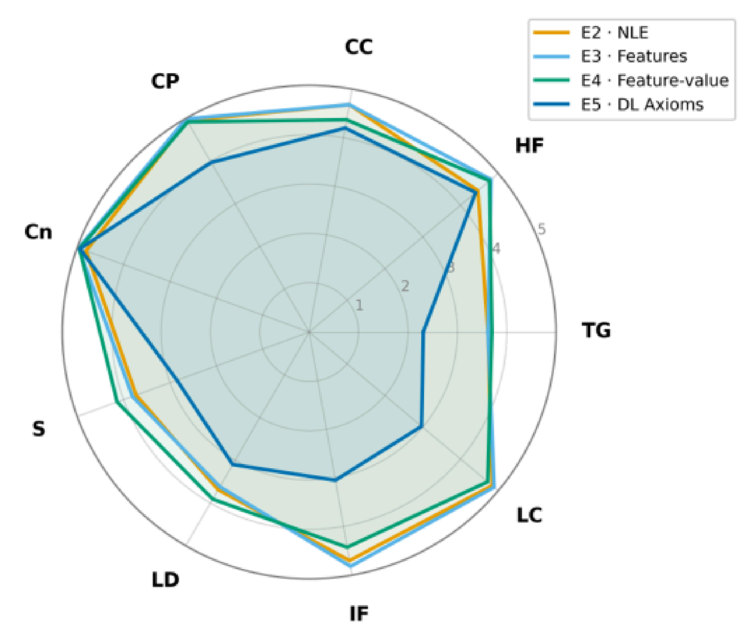

Figure 2. Nine XAI metrics scored by condition by an LLM as-a-judge. **E5 (DL Axioms)** produces a markedly smaller polygon.

DECOUPLING

Accuracy ≠ explanation quality. A model can produce the correct class label while generating a factually incorrect explanation — e.g., correctly identifying a flower species while misreporting the number of petals visible in the image.

08 Discriminativeness as a proxy for correct decisions

Despite the accuracy drop for highly concept-based explanations, **Local Discriminativeness (LD)** emerges as the metric most strongly associated with correct predictions. **When models succeed, they tend to justify their decisions using visual features that effectively separate the target class from alternatives** — suggesting that access to discriminative evidence is closely tied to correctness.

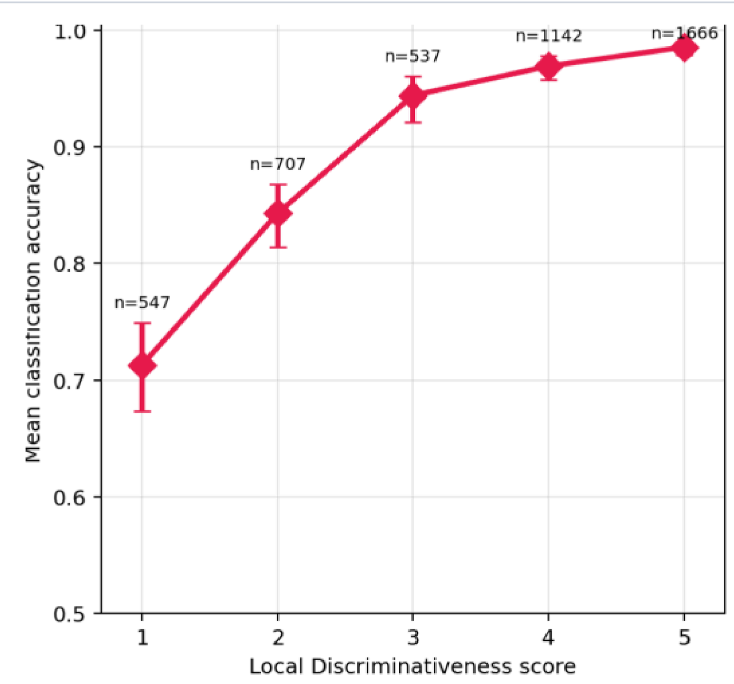

Figure 4. Mean classification accuracy as a function of LD score, pooled across all models, datasets and explanation conditions (E2–E5); error bars: 95% CI over *n* trials.

09 Conclusions

1 Explaining is harder than predicting alone.
Accuracy degrades *monotonically* with explanation rigour (93.8% → 90.1%).

2 Accuracy does not imply explanation quality.
Right label, wrong rationale — explanations need independent evaluation.

3 Discriminativeness as a proxy for correct decisions.
LD is the metric most strongly associated with correct predictions.

4 The instruction-tuning hypothesis.
MLLMs are not optimised to generate formal explanations in tandem with classification.

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